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Come Together, Right Now? An Empirical Study of Collaborations in the Music Industry

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Abstract. Artist collaborations in music have been on the rise, and they tend to produce commercially and critically successful songs. We seek to uncover the effect of these collaborative projects on career trajectories and identify the factors that lift an artist's profile in the short and long term. We develop a theory of collaboration based on the transfer of capital between the collaborating artists that facilitates spillovers across time. To validate the theory, we use weekly radio plays of individual songs across 25 European countries between the years 2011 to 2018, together with a multiattribute Spotify data set of songs and Hofstede's cultural dimensions in relation to artist origins. We create pairs of similar artists who released a collaboration and a solo song in the same week and measure the impact of collaborations based on the difference-in-differences methodology. We find that releasing a collaboration song, in comparison with a solo song, increases the number of plays of an artist in the future by +4.6%. This lift can be broken down into +9.6% for the current song and +7.7% for subsequently released songs, whereas past songs are unaffected. The effect is moderated by the difference in economic, social, and cultural capitals and is significantly larger when one's partner has higher economic and social capital or is highly dissimilar along the cultural dimension. Our theoretical and empirical exploration of such strategic alliances uncovers several underlying mechanisms at play in the success of these pairings and can serve as the basis for future work targeted at prescriptive contributions.

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Keywords: new product development • collaborations • cultural production • music • status hierarchies • complementarities

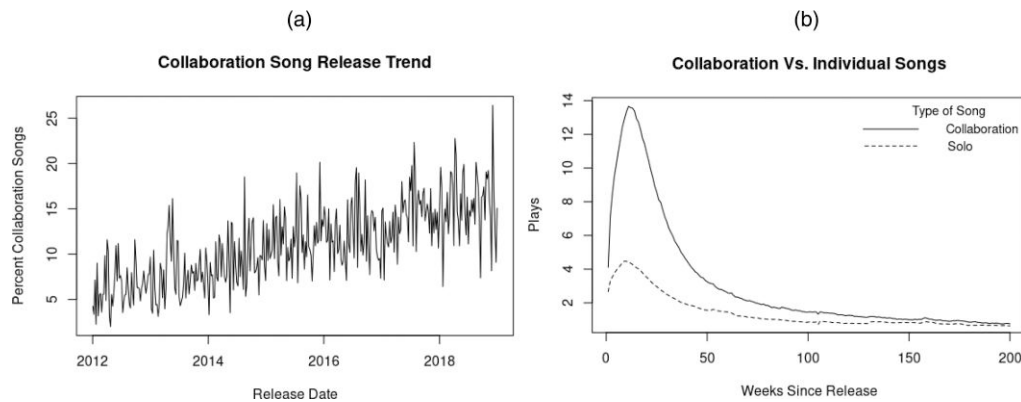
1. Introduction

In commercial music, in which being at the top makes or breaks careers, joining hands with direct competitors and sharing one's recipes for writing hit songs (successful new products) with them may seem unprecedented. And yet collaborations have become increasingly common in recent years in the music industry. Figure 1 shows the percentage of songs that are the outcome of a collaboration (left panel) as well as the related average number of plays per week garnered by a collaboration versus a solo song release (right panel) from our data.

At a first glance, it is obvious that collaboration songs perform overwhelmingly better, garnering more than twice the number of plays per week, on average, compared with solo efforts. This differential in the performance of the two types may be a consequence of the enhanced quality and visibility of the product as a result of the pooling of a diverse set of resources and skills of the participants involved (Uzzi and Spiro 2005,

Gardner et al. 2012, Sommer et al. 2020). Moreover, collaborations may help team members widen their knowledge base and build a network of connections to gain a stronger foothold in the music industry (Cattani and Ferriani 2008, Borjas and Doran 2015). Whereas there are multiple studies that explain the link of collaborations with success (Azoulay et al. 2010, Mannucci 2017, Tan and Netessine 2019), it is not clear whether the mechanisms identified in these works are at play in the context of cultural production. In particular, the literature is silent on the role of complementarities between collaborating parties in the success of the resulting product and the spillover effects on the careers of the artists involved, and this may affect the success of future productions.

Through our work, we advance the theory of collaboration in new product development (NPD) by explicitly recognizing the particular features of cultural creativity. Indeed, the outputs of the creative arts are more than just products and possess the power to alter individuals,

Figure 1. Extent and Impact of Collaborations in Our Radio Plays Data Set

Notes. (a) Percentage collaboration songs by week. (b) Average number of plays per week per song.

societies, and cultures altogether. We choose music as the paradigmatic context that commands massive, global influence, for example, the ever-enduring “mop-top” haircut of The Beatles; the “King of Pop” Michael Jackson’s funeral, which drew in more viewers on televisions across the world than any other death-related event; or 1985’s Live Aid concert that played simultaneously across two continents to raise money for the Ethiopian famine. Frith (1996, p. 109), quite rightly, casts music as an experience, in which it “is both performance and story, describes the social in the individual and the individual in the social, the mind in the body and the body in the mind.” Hence, an analysis involving cultural products requires the consideration of these social influences. We build a theory based on the vigorous framework provided by Bourdieu’s (1986) sociological conceptualization of culture and cultural products. Pairing the existing sociological conventions with economic theory, we validate the theory with granular data on the reach of music to the general audience by considering song-level, weekly plays on the radio across a sample of 125 radio stations in 25 European countries, resulting in 36 million observations that help us capture the dynamics of a wide variety of music.

We intend to quantify the short- and long-term success derived from a collaboration between artists. For this purpose, we carry out our analysis at the artist level so that we can study spillover effects on the rest of the songs in the portfolio of an artist. Specifically, we address the following research questions: (a) What is the impact of a collaborative project, as opposed to a solo effort, on the current song as well as previous and future releases of an artist? (b) How do the complementarities between the collaborators influence this effect? (c) How can artists optimally choose with which peer to collaborate to generate the maximum impact from the collaboration?

To answer these questions, we borrow the basic principles from Bourdieu to develop a theory grounded on

economic, social, and cultural capital. It predicts that the difference in vertical attributes (i.e., economic and social capital) should increase the impact of a collaboration and horizontal differences (i.e., cultural positioning) should have a nonmonotonic impact, in which maximum success is achieved by pairing two artists that are neither too similar nor too distinct. We operationalize these theoretical aspects by combining three unique data sets: a data set of radio plays by song and week across Europe; an exhaustive data set on song attributes, aggregated to the artist level; and Hofstede’s cultural dimensions based on the origins of the artists. We then apply a generalized difference-in-differences (DID) estimation to track the effect of a collaboration on the careers of the artists by matching an artist who collaborates with an almost identical one who releases a solo song at the same moment; we, thus, create pairs of twins with similar career age, number of songs released in the past, popularity, and institutional support. This empirical strategy controls for possible confounders related to artist heterogeneity and potential self-selection concerns, bringing us closer to identifying the causal effect of the collaboration.

Our results show that releasing a collaboration song significantly increases the number of plays an artist receives in the future by 4.6%. In fact, this increase can be traced back to the success of the focal product (9.6% more plays for a collaboration in comparison with a solo effort) as well as the spillover effect on future releases (7.7% more plays compared with not having collaborated in the first place). Interestingly, there is limited spillover into songs from the past, suggesting that collaborations increase an artist’s current capital, which can be applied in future production occasions. Furthermore, we investigate the moderating role of an artist’s hierarchical position in the industry as well as complementarity with the partner. We show that a collaboration revives the career of a prominent artist at the price of removing interest from past songs; on the other

hand, nonelite artists see their future jeopardized for the sake of their past musical profile. We also find that the impact of a collaboration increases when the partner has high levels of economic and social capital and is sufficiently diverse along cultural dimensions and that the strongest influence is driven by economic capital spillovers. Finally, we provide an exploratory guide for artists to choose their potential collaborators for generating maximum impact based on the ever-changing relative positions occupied by themselves and their peers.

We, hence, provide an interdisciplinary perspective in the form of a framework for the quantitative analysis of collaboration in artistic production, applicable not only to music, but also to other forms of NPD projects in cultural production. Moreover, we advance the theoretical understanding of cultural industries by explicitly considering hierarchical and categorical dimensions. Whereas Lena and Pachucki (2013) use their own measures to partially track economic capital in culture, to our knowledge we are the first to conduct a comprehensive consideration of the Bourdiean view and validate it econometrically. From the operations management perspective, our methodology contributes in helping artists to strategically manage their portfolio of songs and make smart choices by identifying the right complementarities that can propel them to long-lasting success.

The rest of the paper is organized as follows. Section 2 provides a review of the relevant literature followed by the development of theory and hypotheses in Section 3. Section 4 provides an explanation about the data and the methodology used, which is followed by the results in Section 5. Finally, a discussion of the robustness checks carried out in support of the main findings is given in Section 6 with Section 7 providing the conclusions and implications. The online appendices provide additional details about the approach, robustness tests, and a rundown of the counterfactual analysis.

2. Literature Review

Our work is first related to the literature in organizational theory and innovation management on teamwork and how it can contribute to successful NPD outcomes. Beyond this general stream of research, our paper especially contributes to studies of cultural production and music in particular, in which the economics and management literatures have gradually explored newer aspects of the processes within creative industries in recent years; see, for example, Eliashberg et al. (2006) or Askin et al. (2019). Finally, there is a nascent literature in operations management devoted to unpacking the mechanisms and best practices to manage cultural goods.

It is well-known that the interactions between the members of an NPD team significantly impact the search process and the quality of the solution produced (Kavadias and Sommer 2009, Sommer et al. 2020). Intrateam

dynamics and its effects on group effort are studied extensively in the organizational theory literature, in which choice and strategic selection are an important part of the discourse. As stated in Gardner et al. (2012), team members who are perceived to be more expert are assumed to possess resources of knowledge and skills that are of value to other team members, not just for achieving collective success, but for their personal accomplishments as well. Gould (2002), in his theoretical model of the emergence of hierarchy in a network, finds that small differences in talent and other kinds of “quality” result in the formation of reputations, which get amplified into large status differences. Milgrom and Roberts (1982) define reputation as a view formed of an individual or organization by another based on past experience, especially as a basis for forecasting future behavior. We borrow from this literature the idea that collaborations are inherently influenced by hierarchical and categorical differences between artists (Mihm et al. 2010, Hutchison-Krupat and Kavadias 2015). Concretely, we use Bourdieu’s (1986) seminal work in cultural sociology to explain the relationship between capital accumulation and the effectiveness of collaborations in the music industry. We focus on the difference between the reputations of the collaborating artists based on performance and social engagement factors to characterize the short- and long-term benefits from the collaboration.

A vast amount of literature is dedicated toward studying the theoretical mechanisms behind the effect of team members’ diversity on the output produced by the teams. Indeed, diversity can be beneficial as it allows for effective recombination of member skills and experiences (Negus 1998). Cattani and Ferriani (2008) posit that heterogeneous teams are more likely to provide fresh, new ideas along with legitimacy and support. Some results are, however, inconclusive: team diversity measured along the dimensions of age, gender, ethnicity, or education level does not appear to be significant (Zhou and Rosini 2015). Using data from motion picture animation, Mannucci (2017) finds that the size and field expertise of a team have an inverted U-shaped relation with the creative output of the respective cross-disciplinary teams. Similar results are observed by Dahlin et al. (2005) and Leonardi (2011), among others. The notion that is resonated across all these works, as Milliken and Martins (1996, p. 403) point out, is that “... (Diversity) appears to be a double-edged sword, increasing the opportunity for creativity as well as the likelihood that group members will be dissatisfied and fail to identify with the group.” In our work, we consider the differential in artists’ categorical positioning in the market and study the impact of complementarities on the success of a collaboration.

In the domain of cultural products (see Candia et al. 2019 for a recent description), studies of collaborative NPD projects are scarce. In the movie industry, Elberse (2007) singles out the effect of star actors on the

performance of movies at the box office and on the careers of the costars. Collaborations in music, in which competing artists have the luxury to collaborate with their peers for a short period of time and reap the benefits of this joint work for an extended period throughout their respective careers, remain, however, poorly understood. Uzzi and Spiro (2005), Smith (2006), and Mauskapf et al. (2017) come the closest to our paper, in which the innovation production potential of the artists and their position in the general network are analyzed, within Broadway musicals and commercial music. Uzzi and Spiro (2005) find an inverted U-shaped relationship between innovativeness and the embeddedness of an artist within the Broadway scene network. In contrast, Mauskapf et al. (2017) find that culturally and geographically embedded artists score higher on their innovation index. Smith (2006), more descriptive in nature, carries out network analysis to study the community structure within the rap genre. These works, thus, focus on social network drivers of artistic innovation. In comparison, we look at short-term, two-way collaborations between artists, quantify its economic impact on the participating artists' careers, and identify the moderating factors that maximize this effect. These moderating factors not only include the difference in the embeddedness of the collaborators; we also look at distances in economic and cultural aspects of the artists to provide a comprehensive look at the overall mechanism.

The operations management literature is also slowly turning toward developing insights to better manage cultural products. For instance, Das Gupta et al. (2016) and Tereyağoglu et al. (2018) use reference effects to model optimal sequencing and duration of activities and construct pricing policies, respectively; Martínez-de Albéniz and Valdivia (2019) examine the impact of exhibition content on visitor volume to optimize museum programming; Martínez-de-Albéniz et al. (2020) study restaurant consumer trajectories and provide optimal paths to maximize experienced utility. Studies focused on people-centric operations, such as Huckman et al. (2009), Huckman and Staats (2011), Staats (2012), Ramachandran et al. (2017), Valentine et al. (2019), Tan and Netessine (2019), and Chan et al. (2020), address the effect of fluid team collaborations and project team member selection on performance metrics. We extend these literature streams to explore the underlying relationships and provide a descriptive mapping of the mechanisms at play for the effect of collaborations on the careers of music artists (Roels and Staats 2021).

3. A Theory for Artistic Collaboration

In this section, we first lay down the instrumental preliminaries and develop our theory on artistic collaboration. Based on it, we hypothesize the impact of a collaboration

on the careers of the music artists and discuss the implications of the collaborator partner choice.

3.1. Artist Standing

Success in music depends on the validation provided by others and numerous other factors over which the artists often do not have any control (Bourdieu 1993). For a close-knit community of elite music artists, entry has several process- and performance-related barriers. Faulkner (1983) suggests that artists with excellent recognition possess an advantage in terms of opportunities over those without; Lena and Pachucki (2013) state that the art worlds operate on a currency earned through peer esteem and financial rewards. This paves a way for the following constructs based on Bourdieu's (1986) framework, which help us operationalize the dynamics of this currency gain and loss.

3.1.1. Economic Capital. At any point in time, an artist is either contractually bound to a record label or may be an independent song producer. The record labels invest in the production, distribution, and marketing of the songs written by their artists. In such a scenario, the performance of the previous songs (measured through the sales of their albums, tickets sold at their concerts, etc.) justify the decision of the record labels to continue investing more in the future projects of the artists. For independent artists, the success of a song aids them in putting together a portfolio that attracts the big record labels to "sign them on" (Negus 1992).

The music industry is undeniably run by the economics of the system. This is termed "large scale cultural production" by Lena and Pachucki (2013) and gives rise to a principle of hierarchization that prizes economic capital above all. Consistently producing well-distributed songs earns an artist a reputation of being "successful" and inevitably makes more resources available for future NPD projects. In our paper, the number of radio plays per song garnered by an artist in the recent past is used as a representation of the economic capital, which we explain in more detail in the subsequent sections. Given its temporal nature, this fluctuating measure captures the contemporaneous economic capital possessed by the artists, thereby providing an indication of their current status in the hierarchy.

3.1.2. Social Capital. Sociology conceptualizes the dynamics between an individual and the community as a collective sense of identity in the form of shared beliefs, behaviors, likes or dislikes, which are continually updated through interactions between individuals. In this view, the network made of links between individuals determines the power of influence of everyone in the community. An artist can, thus, gain higher social capital by being connected to many individuals in the music industry (not only other artists but other roles, such as

songwriters, producers, or technicians) or, alternatively, by being connected to a select few individuals who hold a central position within the industry (Barabási and Albert 1999). These connections can be built over time, or an artist may simply inherit them by being associated with a central entity in the community in the form of either another big artist or a big record label. For example, Dr. Dre has earned a significant reputation for himself in making hit songs. However, in his capacity as a record producer, he has also earned a name for having impeccable taste in scouting the right talent and providing them with opportunities—case in point being Eminem and 50 Cent. As a result, to be vouched for by someone as influential as Dr. Dre can, by default, provide an artist with an inflated reputation. Therefore, the uncertainty about someone's ability can be resolved by inferring it from the identities of their affiliates (Blau 1955).

One popular measure for the social capital enjoyed by an artist at any given time is degree centrality (Freeman 1978), in our case, measured as the number of direct connections an artist has through previous collaborations in the extended network. Cowan et al. (2007) substantiate the relevance of this measure in the context of collaborative NPD, which is also supported by Pontikes et al. (2010) and Mauskopf et al. (2017) within the specific domain of cultural production. We study here how this measure of social capital moderates the impact of collaborations.

3.1.3. Cultural Positioning. Whereas economic and social capitals are of a vertical nature (the capital of any two artists can be compared), cultural positioning has a distinctive horizontal nature. Bourdieu conceptualizes cultural capital as a bestowal of legitimacy from the society onto an individual. It takes the form of recognition, that is, a collection of interactions between the individual's positioning and the environment's response to that position statement. In other words, individual attributes, acquired or inherited (often simply by being part of a certain social group), grant the artists a certain reputation. Nowhere is this more obvious than in rap, in which an artists' local allegiances and territorial identities stand out (Armstrong 2004). Moreover, artistic inclinations developed through previous works help artists create niche areas for themselves, thereby earning an expertise in corresponding fields (Askin and Mauskopf 2020). Hence, cultural positioning is instrumental in the categorization of the music industry as opposed to the hierarchical nature of economic and social capitals.

Looking through an ethnographic lens, we assess the cultural characteristics of an artist as the relative position the artist occupies in the musical community along the following two dimensions:

- **Artistic attributes:** The position acquired by the artists by creating a niche area for themselves grants

them legitimacy. They are perceived as experts of that particular stream of music (DiMaggio 1987, Askin and Mauskopf 2020). For example, Shakira, Colombian by birth, released her first song at the age of 13. Through a consistent stream of releases, which are primarily Latin pop, she has carved out a corresponding image for herself in the music industry that reflects this uniqueness about her. Hence, Shakira's artistic attributes are measured by the musical attributes of her songs, such as danceability or other musical features (see Online Table 5 for details).

- **Origin-culture attributes:** We posit that the origin of an artist plays an important role in how the artist is perceived by both audiences and peers in the music industry. When Shakira includes "Let me see you move like you come from Colombia" in the lyrics of her hit song "Hips Don't Lie," she is vying to be identified by her Colombian identity and maintain an authenticity that contributes toward creating a stable image for herself (Negus 1992, Armstrong 2004). We measure this perceived image of an artist through Hofstede's (1984) four cultural dimensions without commenting on how much an individual artist conforms to the stereotype associated with the culture of origin (see Online Table 6 for details).

Section A in the online appendix provides an illustration of these Bourdiean capitals for the career of Shakira and a few of her peers in the music industry.

3.2. What is a Successful Collaboration?

A collaboration occurs when two independent artists or groups of artists come together to write, record, and promote a single song or an album. These collaborative NPD projects are identifiable by the signature of the song, in which one artist is followed by a second "featuring" artist. Because both artists are credited for the song and we do not observe the breakdown of the corresponding monetary gains, we assume that they make symmetric contributions toward the project. As do Uzzi and Spiro (2005), we focus our analysis on the collaboration between a pair of core artists and do not consider the extended crew, which includes technicians and specialists whose services are hired toward producing the song. Note that these may have made artistic contributions toward the project, but because their names are not advertised, their reputations, substantiated through the Bourdiean view in our theory, should have little effect on the performance of the song in the public domain, and hence, we remove them from our analysis. Notwithstanding, the institutional support provided by the record labels to their artists in processes ranging from music production to marketing might have an effect on subsequent success levels. We, thus, take the necessary measures to track these potential external contributions to a song in our econometric approach.

The focus of this paper is to measure the impact of a collaboration on a commercial success metric. Specifically, we measure the success of a song (new product) as the number of plays per week garnered by that song on the radio, the quintessential cultural intermediary for music (Coase 1979, Musgrave 2017). This measurement, which provides the most tangible commercial recognition for the song and, hence, is of the utmost importance for the subsistence of the artist, is in line with the standard metrics used for assessing the product performance in the NPD literature (Cohen et al. 2000). Our approach, however, steers clear of assessing the “true quality” of the artistic product generated. The notion of objective evaluation of the arts remains, as Bleich (1975) puts it, “ludicrous.” In addition to interpretive knowledge, which is constructed from the underlying characteristics of the art and the uncontrolled experiences of the consumers, their ever-evolving tastes, and preferences make the true quality of music ambiguous (Lamont and Molnár 2002).

3.3. Why Collaborate?

As described in the introduction, the music industry is very competitive: as competition in recent years has become more intense, an artist not only struggles for creating new content, but is also compelled to reach higher levels of acclaim and prestige for survival (Musgrave 2017). Collaborating with a suitable partner is an effective way for the focal artist to get support during these struggles for a short period of time (Cowan et al. 2007, Bhaskaran and Krishnan 2009) on several dimensions that we detail next.

3.3.1. Spillover Effects. Joining forces with competitors not only reduces the direct competition that an artist may face, but there are positive spillover effects beyond the short-term, economic realm. Note that outcomes must be compared with an appropriate counterfactual, that is, that of releasing a solo song.

On production economics, Borjas and Doran (2015) identify collaboration networks to be an important dimension for knowledge spillover. Repeated collaborations help build network resources that serve as channels of valuable information exchange and reduces search costs for finding future collaborators or support teams such as technicians or managers (Sommer et al. 2020). In addition, collaborations tend to have a transformative effect on the participants, providing them with opportunities to assimilate their partners’ skills and apply them in their own works to reap long-term benefits (Chan et al. 2018, Liu et al. 2018). Creatively speaking, cross-disciplinary collaborative NPD projects enable interactions between people with different sets of contacts, skills, information, and experiences, which facilitates production of novel work (Guimera et al. 2005).

Moreover, collaborations also have a strong influence on distribution economics. One underlying motivation for an artist to collaborate is to extend the artist’s “base market” (Negus 1992) and make the music available to that artist’s partner’s regular listeners. This newly found exposure to a potentially larger audience catapults the collaboration project to levels of success that are unattainable for the artists on their own. We theorize that the expansion of the follower base goes beyond the collaboration project: it makes a lasting impact on the careers of both the artists to the extent that their previous and future individual projects also benefit from this pairing. Indeed, having been exposed to a new strata of audience, which was previously unapproachable, the curiosity of these audiences leads them to consume other works by the newly discovered artist (DiMaggio 1987, Goldberg et al. 2016). Consequently, the utility of playing the collaboration song or past and future individual releases of an artist increases as a direct consequence of these two artists teaming up (Becker 1991). Hence, we hypothesize the following.

Hypothesis 1. *A collaborative project has a positive effect on the overall performance of songs from an artist in comparison with a solo project.*

Hypothesis 2. *A collaborative project has a positive effect on the performance of songs released in the past by an artist in comparison with a solo project.*

Hypothesis 3. *A collaborative project has a positive effect on the performance of songs released in the future by an artist in comparison with a solo project.*

Note that we do not include a hypothesis about the effect on the collaboration song given that Figure 1 already shows that it performs better than a solo song.

3.3.2. Ceiling Effect. Whereas the effect of a collaboration may be omnipresent, we propose that its intensity varies according to the focal artist’s status in the music industry. Artists who command high sales, generate high revenues, or—as per our measure of economic capital—garner a large number of plays on radios are considered to be the high-status elites (Azoulay et al. 2010, Pontikes et al. 2010, Lena and Pachucki 2013). We expect them to be subject to the Matthew effect (Merton 1968), in which the perceived quality of the elite artists’ work is always positively biased and receives undue recognition, leading to broader audiences. At the same time, they have less to gain in terms of greater access to production and distribution resources. This is called the Mark effect (Bothner et al. 2011), which suggests that there is a redistributive intervention in which nonelite artists enjoy a larger benefit from the collaboration project. In other words, we theorize the presence of a ceiling effect, whereby there exist diminishing returns in the benefits from a collaboration for artists

who are already elite in the hierarchical structures of the music industry, in comparison with the collaborations of the nonelites from the lower echelons (Azoulay et al. 2014).

Hypothesis 4. *A collaborative project has a larger positive effect on the performance of nonelite artists in comparison with the elites.*

3.4. With Whom to Collaborate?

Whereas a heterogeneous mixture of skills generates the best possible output (Tan and Netessine 2019), complementarity of these skills and resources between the collaborators is even more crucial (Sinha and Cusumano 1991). Our theory so far suggests that a collaboration in itself provides the career of the focal artist a significant boost, but going a step further, if the selection of the collaborator is carried out strategically, the focal artist can effectively control the magnitude of this boost received from the partnered project. Kaplan and Vakili (2015) and Huckman and Staats (2011) point out the “double-edged sword” of combining knowledge from extremely diverse or extremely similar domains in the creation of an innovation and NPD. We propose that the efficacy of the collaboration in the music industry is moderated by the complementarity of skills and resources. Given the tacit nature of music, economic, social, and cultural positions in the industry provide all the elements to guide artists in their hunt for the right collaborators (Levin and Cross 2004). Accordingly, the extent to which collaborating artists differ from each other along these capitals determines how successful their collaborative project becomes and the spillover effects it has on their respective careers in the long run.

3.4.1. Difference in Economic Capital. In any dyad, there is an artist who is relatively more successful in terms of the distribution of previous songs, making the other one less successful: a form of status order (Lena and Pachucki 2013). This duality can be a reflection of the intrinsic musical quality of the two artists or merely the fact that one artist has been in the music industry for much longer than the other.

It is crucial to identify the two mechanisms driving this moderator. On the one hand, there is a direct effect observed on the focal artist’s career when the artist collaborates with someone more successful—that is, receiving more plays—than the artist (Jin et al. 2019), which brings wider market access and larger resources to the focal artist. Hence, in this mechanism, the focal artist always wants to be associated with a more successful artist (Gould 2002). By such association, the artist with lower economic capital may also gain higher status (Thibaut and Kelley 1959). On the other hand, as in Hypothesis 4, we anticipate that elite artists have less to gain from a collaboration, suggesting that the impact of the

collaboration should decrease in the focal artist’s economic capital. Taking these two forces into account, we theorize that there is a transfer of economic capital from the high-status partner to the low-status artist, by which they experience a negative and positive boost in their plays, respectively. Specifically, we expect this moderation effect to be captured by the difference of economic capital (e.g., proportion of the plays as described later in Section 4) between the partner and the focal artist, which should reduce the impact of the collaboration on the focal artist’s career.

Hypothesis 5. *The effect of a collaborative project decreases in the difference in economic capital between the focal artist and the collaborator.*

3.4.2. Difference in Social Capital. Synchronous with the argument for economic capital, social capital also generates a dynamic of imbalance in a dyad. Singh (2005) and Valentine et al. (2019) empirically show how possessing higher social capital can have a positive impact on the resource exchange and combination. We theorize that the focal artist can leverage the new-found increased influence through these connections to gain a stronger foothold in the industry, consequently making more and better opportunities available to the artist in the future (Van den Bulte and Moenaert 1998). By collaborating with another artist, the artist also gains indirect access to the collaborator’s collaborators—a socially central artist to the peripheral partner’s non-mainstream collaborators and the peripheral artist to the well-connected partner’s mainstream network (Cattani and Ferriani 2008, Ductor 2015, Liu et al. 2018). Similar to economic capital, there is a transfer of social capital, which is especially beneficial to the focal artist if the partner is highly central in the music industry, resulting in the much larger effect of the collaboration than if the artist’s partner is socially isolated. Again, we expect the moderation effect to be driven by the difference in social capital.

Hypothesis 6. *The effect of a collaborative project decreases in the difference in social capital between the focal artist and the collaborator.*

3.4.3. Distance in Cultural Positioning. A dyad of diversely skilled artists can generate higher creativity through access to a wider pool of knowledge (Uzzi and Spiro 2005), whereas a culturally diverse dyad can approach problems with a wider set of strategies (Kavadias and Sommer 2009, Sommer et al. 2020). When the artists in a dyad are too far apart from each other along the artistic and cultural attributes, peculiar products may result from their collaboration. Additionally, the regular listeners of both the artists may perceive the collaboration to be a match of incompatible styles. Such a radically innovative product, which ex ante generates

apprehension for its artistic direction, may turn out to be too difficult to interpret and may be considered “ahead of its time” and, in general, be ignored by the audiences because of a lack of adopters (Chao and Kavadias 2008, Uzzi et al. 2013). For example, when The Beatles used Pt. Ravi Shankar’s distinctive Indian music in the track “Within You Without You,” part of their *Sgt. Pepper’s Lonely Hearts Club Band* album, the track failed to gain much attention because the “base markets” of both music producers failed to connect with the marriage of two extremely different musical styles. Moreover, the process of creation itself may be plagued by insurmountable disagreements, which result in delays and suboptimal compromises (Dahlin et al. 2005, Kavadias and Sommer 2009, Huckman and Staats 2011). On the other hand, if the artists are too similar, their collaboration effort yields an uninspiring outcome that fails to attract new audiences out of their regular domain. By collaborating with an appropriately complementary artist and relying on the partner’s legitimacy in a subdomain alien to the artist, the focal artist can cast the collaboration project to be an interesting prospect to which to listen for the artist’s audiences, increasing the probability of success of the focal song and ensuring positive spillovers on the artist’s remaining profile (Gardner et al. 2012, Kim et al. 2020). In other words, there are two opposing forces that influence the impact of the collaboration: when the distance in dyad cultural positions is too small or too large, the effect of collaboration is less salient, whereas intermediate distances provide a sweet spot that maximizes the performance of the collaboration. We, thus, hypothesize that the effect of the collaboration follows an inverted U-shaped relationship with respect to the artistic and cultural distances between the collaborating artists, in line with Dahlin et al. (2005) and Mannucci (2017). Whereas the hypothesis is established before in other contexts, we seek to extend its validity to the music domain.

Hypothesis 7. *The effect of a collaborative project follows an inverted U-shape with respect to the distance in cultural positioning between the focal artist and the collaborator.*

4. Data and Methodology

4.1. Original Data

We use a unique, large data set assembled by combining three data sources that covers all the aspects of song performance as well as artist attributes. It has a panel structure by which an observation provides the number of plays on a given radio station in a given week for a given song.

First, we obtained a primary data set that is a log of the songs played by 125 radio stations across 25 countries in Europe from January 2011 to December 2018. The data are collected by BMAT, a music identification firm that continually scans radio programs and, via

pattern recognition, identifies the song that is being played so that royalty agencies can correctly pay artists their dues. A typical observation is shown in Online Table 4 in Section B.1. Whereas we keep the data from 2011 to create measures of artist attributes, we focus our performance analysis on all the songs released from 2012 onward. We aggregate the plays for each song at a weekly level, normalize them to remove song life cycle effects (see Online Appendix B.2 for details), and narrow down our data set to the 90th percentile artists, that is, those artists with more than 625 plays across the years. By doing this, we focus on the prominent members of the music industry for whom reputation and status are relevant and our Bourdiean view applies. This filtering process yields a final sample of 7,370 artists with 95,283 songs, which account for 91.0% of the total plays registered. Note that, in our robustness checks, we widen this subset of artists to the 40th percentile, obtaining qualitatively similar results, albeit with different strengths depending on the status of the artist. A song that has “featuring” or any of its variants in the title or artist credit is identified as a collaboration: 3,518 artists collaborate to make 6,065 songs (6.3% of sample songs) accounting for 13.7% of the total plays (15.1% of sample plays).

Second, because of recent advances in machine learning techniques, a more fine-grained and feature-based representation of music has become available (Askin and Mauskopf 2020). We resort to the song-level audio features designed by Echo Nest, now a part of Spotify, so the data are available through the Spotify application programming interface. It provides a quantified representation of both technical and nontechnical attributes of songs, detailed in Online Table 5 in Section B.3. We aggregate these to a time-varying artist-level series such that, for an artist i at any time $t \in [1, T]$ with S_{it} songs in the market, an audio feature F_{it} is

$$F_{it} = \frac{1}{S_{it}} \sum_{s=1}^{S_{it}} f_{ist},$$

where f_{ist} is the audio feature of a song s by artist i . Accordingly, we get audio features for 3,894 artists that are used to measure the artistic distance between the collaborators. For the remaining 3,476 artists in the sample, we have no information about their artistic attributes. Online Figure 4 in Section B.3 shows how these audio features accurately track the relative positions occupied by the artists in the musical sphere.

Third, we gather cultural information about the artists. We use web crawlers on Wikipedia to gather data about the origin of 4,514 artists. We use this in conjunction with Hofstede’s cultural dimensions (Hofstede 1984) to determine cultural distance between artists (see Online Appendix B.4 for details).

4.2. Econometric Strategy

Our aim is to capture the effect of a collaboration on the overall profile of the participating artists. Because we are interested in spillover effects across the songs in the artist's portfolio, this necessitates an analysis at the artist level. We tease out the effect of a collaboration on the plays garnered by the focal artist, through an analysis of the pre and post time periods with respect to the release date of that song. Of course, to properly measure the impact, we need to build an appropriate counterfactual, which takes the form of the release of a solo song. This suggests that we need to find a "twin" artist that followed the counterfactual strategy and necessitates the development of a matching approach, described next.

4.2.1. Matching. The decision of collaboration is likely to be confounded by the focal artist's contemporaneous career trajectory. For instance, an artist with a very high number of plays on the radio and who has collaborated several times in the recent past is more likely to collaborate than someone who has not done so.

We, hence, need to account for the endogenous decision of collaboration by carrying out a specialized matching procedure whereby a pair of artists is created based on the following observable factors: (a) release date of the song t , (b) career age ($CareerAge_{it}$: the number of weeks since the first song of artist i from Spotify), (c) number of individual songs released by the artist in a time window of one year prior to t ($IndivSongs_{it}$), (d) number of collaboration songs released by the artist in a time window of one year prior to t ($CollabSongs_{it}$), synonymous to their social capital, and (e) the log of the cumulative number of plays garnered by the artist per song in a time window of one year prior to t :

$$CumPlays_{it} = \log \left(1 + \frac{1}{S_{it}} \sum_{s=1}^{S_{it}} \sum_{t'=t-52}^{t-1} Plays_{st'} \right).$$

Note that $CumPlays_{it}$ is independent of the career age and the number of songs the focal artist has and, hence, should indicate the artist's contemporaneous economic capital in the market.

Whereas the aforementioned strategy is directed toward controlling for the artist's endogenous choice of deciding to collaborate with one of the artist's peers, the radio stations' choice of playing music of one artist over the others' can potentially bias the inferences from our analysis. Specifically, record labels have been known to actively promote the songs of their artists on radio stations by investing monetary resources (which was made customary to be disclosed in public records in the United States as early as 1934 through section 317 of the Communications Act)—a phenomenon more commonly known as "payola" (Coase 1979). To control for this and in line with the common practice in studies related to analyses on music performance, such as Askin et al. (2019) and

Askin and Mauskapf (2020), we add two additional binary parameters of (f) $PrevBigLabel_{it}$ and (g) $BigLabel_{it}$. Here, the first variable takes a value of one if the artist was associated with one of the Big 5 labels (Sony, Warner, Universal, BMG, and EMI) prior to releasing the new song, and the second variable takes a value of one if the new song is released with one of the Big 5 labels (Bishop 2004). These factors help us control for the quality of the behind-the-curtains support that the artists get when producing their new songs, and it allows us to partially capture the external contributions to a song beyond that of the artist.

We say artist i is treated if the artist releases a collaboration song at time t , whereas the match artist i' is the artist's control if the artist releases an individual song at the same time, that is, identical factor (a). To ensure that we capture the effect of the current collaboration on the outcome variable and not other past releases, we force the pair of artists to have an equal number of individual and collaboration releases during one year prior to t , that is, identical factors (b) and (c). To ensure that the support they get from their producers, monetary and otherwise, is consistent, we get an exact match on factors (f) and (g). Finally, we carry out propensity score matching with a narrow caliper of 0.5 to get a single match for the treated artist (Craig et al. 2016, Liu et al. 2018), that is, neighboring factors (d) and (e) without replacement. This process yields 2,116 matches; using calipers from 2 (less strict) to 0.15 (more strict) gives us matches ranging from 2,148 to 1,771. The statistics for the matched data given in Online Appendix B.7 indicate the effective balance between the treated and control groups. Note that including $IndivSongs_{it}$ and $CollabSongs_{it}$ in our matching procedure also creates an effective balance for the time lag (in weeks) between the focal release and the corresponding previous release represented in $TimeSinceLast_{it}$. Examples of a few matched pairs are shown in Table 1.

With our matching procedure, a pair of matched artists that share similar values for factors (a)–(g) are twins in terms of past song performance. Moreover, we check that there is no anticipation of the treatment reflected in the plays on the radio in the pretreatment period; see Online Appendix E.2. As a result, the difference in future performance can be attributed to the collaboration event. A multitude of alternate matching parameters are also utilized as robustness checks, which we elaborate on in Section 6.3.

4.3. Generalized Difference-in-Differences

4.3.1. Base Model. The decision to collaborate taken by the artists in our data set, which is considered as a treatment in our context, is not simultaneous, which calls for the use of a generalized DID approach (Cui et al. 2019, Calvo et al. 2020). Two distinct time lines are used in our analysis. For a given pair of treated/control artists k , t_k denotes the time at which the matched pair

Table 1. Examples of Matched Pairs

Release date	Artist	Track	Collab.	Career Age _{it}	Cum Plays _{it}	Indiv Songs _{it}	Collab Songs _{it}	Prev BigLabel _{it}	BigLabel _{it}
2013-06-10	Drake	Right Here (with Justin Bieber)	1	195	0.23	2	0	1	1
2013-06-10	Ycare	Pourvu Que Tu Viennes	0	206	0.11	2	0	1	1
2013-10-28	Iggy Azalea	Change Your Life Feat. T.I.	1	42	0.81	3	0	1	1
2013-10-28	Chvrches	Lies	0	42	0.83	3	0	1	1
2015-03-02	Alesso	Cool Feat. Roy English	1	246	1.84	1	4	1	1
2015-03-02	Jason Derulo	Want To Want Me	0	276	1.93	1	4	1	1
2016-10-24	Duke Dumont	I Got U Feat. Jax Jones	1	483	2.09	4	0	1	1
2016-10-24	Lady Gaga	A-YO	0	459	1.31	4	0	1	1
2017-05-01	Michael Carreira	Ya Ya Ya Feat. Sebastian Yatra	1	548	0.78	2	0	1	1
2017-05-01	Miley Cyrus	Malibu	0	538	0.82	2	0	1	1

releases its new song. Time t represents the absolute time, whereas $\tau = t - t_k$ is the number of weeks since the new song was released. Hence, $\tau < 0$ becomes the pretreatment period, and $\tau > 0$ becomes the posttreatment period.

We estimate the average effect of the release of a collaboration song on the number of plays garnered by the artists through the following DID model specification:

$$\log(Y_{ik\tau}) = \alpha_1 \text{AFTER}_{\tau} + \alpha_2 C_{ik} + \beta \text{AFTER}_{\tau} \times C_{ik} + \gamma Z_{it} + \Omega_k + \mu_t + \epsilon_{ik\tau}. \quad (1)$$

In Equation (1), the dependent variable can take different forms $Y_{ik\tau} \in [\text{TotalPlays}_{ik\tau}, \text{FocalSongPlays}_{ik\tau}, \text{PastSongPlays}_{ik\tau}, \text{FutureSongPlays}_{ik\tau}]$, where

a. $\text{TotalPlays}_{ik\tau}$ is the average number of plays per week per song, garnered by artist i , within the pair k , across all the songs on the artist's profile τ weeks after the song's release.

b. $\text{FocalSongPlays}_{ik\tau}$ is the number of plays per week garnered by the focal song by artist i at τ .

c. $\text{PastSongPlays}_{ik\tau}$ is the average number of plays per week per song, across the songs released by artist i prior to t_k , at τ .

d. $\text{FutureSongPlays}_{ik\tau}$ is the average number of plays per week per song across the songs released by artist i after t_k at τ .

Hence, the effect of a collaboration on the overall trajectory of an artist is identified through the analysis of $\text{TotalPlays}_{ik\tau}$, whereas the spillover effect of a collaboration on the plays of the focal artist's other songs is inspected through $\text{PastSongPlays}_{ik\tau}$ and $\text{FutureSongPlays}_{ik\tau}$. We also measure the effect of a song being a collaboration on the number of plays it garners in $\text{FocalSongPlays}_{ik\tau}$. We use a log transformation of the dependent variables to control for the skewness in which we add a small number to avoid problems with zero values, that is, the dependent variable is $\log(0.1 + \cdot)$.

The independent variables in Equation (1) use the classic DID structure. AFTER_{τ} is a binary indicator of

the posttreatment period such that $(\text{AFTER}_{\tau} | \tau \geq 1) = 1$. $C_{ik} = 1$ is the formal indicator for the focal artist i releasing a collaboration song in pair k , whereas $C_{ik} = 0$ indicates the control artist. Z_{it} includes two time-varying control variables for the artist: career age of the artist at τ and dummy variables to control for the release of a new song during $\tau > 1$. Furthermore, we add pair- and year-level time fixed effects to ensure enough degrees of freedom (Ω_k and μ_t , respectively) and control for time-invariant factors. Finally, because the unit of randomization is at each observation level, standard errors are clustered at the focal artist and pair levels, that is, errors of all observations sharing the same artist i and the same pair k , but not τ . Note that using robust errors does not change our results, that is, significance of coefficients is qualitatively similar.

We further divide our sample into four quartiles based on CumPlays_{it} (which is used in the matching process), assuming it to be the proxy for the contemporaneous position of an artist in the music industry. For example, Cardi B, the American rapper, who was still on the fringes of the music industry during the mid 2010s, is rightly placed in the second quartile of our analysis. Artists with CumPlays_{it} in the first quartile are considered to be nonelite artists, whereas those in the fourth quartile are the elites. Boudreau et al. (2011) and Ductor (2015) use a similar kind of stratified analysis. Our theory predicts that the then-nonelite Cardi B would stand to gain tremendously from collaborating, which, as of the early 2020s, has only turned out to be true.

We also consider an alternate plausible approach as followed by Bell et al. (2018), in which individual (artist) fixed effects are used. In our case, it reduces the number of degrees of freedom significantly and generates results qualitatively similar to those included in the paper.

4.3.2. Model with Complementarities. As in Azoulay et al. (2014), Calvo et al. (2020), and Cui et al. (2019), we study the moderating effect of differences in economic capital, social capital, and cultural positioning between

the focal artist i and the collaborator on the career trajectory of the focal artist through a modified version of the previously defined DID specification:

$$\log(Y_{ikt}) = \alpha_1 \text{AFTER}_t + \alpha_2 C_{ik} + \beta \text{AFTER}_t \times C_{ik} + \theta C_{ik} \times M_{ikt} + \lambda \text{AFTER}_t \times C_{ik} \times M_{ikt} + \gamma Z_{it} + \Omega_k + \mu_t + \epsilon_{ikt}. \quad (2)$$

In comparison with Equation (1), we make two modifications. We expand the control Z_{it} to include artist quartile position so as to control for focal artist status. We also add four moderators $M_{ikt} \in [\Delta \text{CumPlays}_{ikt}, \Delta \text{DegreeCentrality}_{ikt}, \Delta \text{Art}_{ikt}, \Delta \text{Origin}_{ikt}]$. The first three moderators vary over time, whereas the last one stays constant:

- $\Delta \text{CumPlays}_{ikt}$ is the difference between the log of total plays per song per week of the focal artist i ($\text{Focal_CumPlays}_{kt}$) and the collaborating partner ($\text{Partner_CumPlays}_{kt}$) in pair k . It captures the moderating effect of the difference in their economic capitals given by $\Delta \text{CumPlays}_{ikt} = \log(\text{Focal_CumPlays}_{kt}) - \log(\text{Partner_CumPlays}_{kt})$.

- $\Delta \text{DegreeCentrality}_{ikt}$ is the difference between the log of degree centrality of the focal artist i ($\text{Focal_DegreeCentrality}_{kt}$) and the partner ($\text{Partner_DegreeCentrality}_{kt}$) in pair k . It, thus, tracks the moderating effect of the differences in their social capitals through $\Delta \text{DegreeCentrality}_{ikt} = \log(\text{Focal_DegreeCentrality}_{kt}) - \log(\text{Partner_DegreeCentrality}_{kt})$ (details about our estimation of the degree centrality are provided in Online Appendix B.5).

- ΔArt_{ikt} is one minus the cosine similarity between the artistic attributes of the collaborating artists in pair k given by Focal_Art_{kt} and Partner_Art_{kt} , thereby providing a distance value between zero and one. Hence, Partner_Art_{kt} , thereby providing a distance value between 0 and 1. Hence, $\Delta \text{Art}_{kt} = 1 - \frac{\langle \text{Focal_Art}_{kt}, \text{Partner_Art}_{kt} \rangle}{\|\text{Focal_Art}_{kt}\| \times \|\text{Partner_Art}_{kt}\|}$.

- $\Delta \text{Origin}_{ikt}$ is one minus the cosine similarity between Hofstede's four cultural dimensions of the collaborating artists in pair k given by Focal_Origin_{kt} and $\text{Partner_Origin}_{kt}$, thereby providing a distance value between zero and one. Hence, $\Delta \text{Origin}_{ikt} = 1 - \frac{\langle \text{Focal_Origin}_{kt}, \text{Partner_Origin}_{kt} \rangle}{\|\text{Focal_Origin}_{kt}\| \times \|\text{Partner_Origin}_{kt}\|}$.

For these four moderators, we add linear as well as quadratic effects (Haans et al. 2016, Mannucci 2017) without interactions. We set these values to zero when the focal artist releases an individual song, which allows us to remove the term $\text{AFTER}_t \times M_{ikt}$, which is identical to $\text{AFTER}_t \times C_{ik} \times M_{ikt}$ (see Ductor 2015 for a similar approach). Descriptive statistics for these variables are given in Online Appendix B.6. Because these moderators are not available for all artists (recall that artistic and origin attributes are only obtained for a subset of artists), the sample size is reduced from 4,232 to 1,148 artists. The descriptive statistics for these variables are provided in Online Appendix B.6.

5. Results

5.1. The Overall Effect of a Collaboration

Let us first look at the overall effect of a collaborative project on the total plays. Model 1, shown in Table 2, suggests that releasing a collaboration song gives an additional +4.6% ($= e^{0.045} - 1$, p -value < 0.1) boost to the number of plays of an artist in comparison with the release of an individual song. Hence, a collaboration has a small but positive impact (comparatively weak signal) on the career of an artist, validating Hypothesis 1.

However, this effect is not homogeneous across all types of artists. Table 2 shows the model estimates for the subsamples made of first and fourth artist quartiles, whereas Figure 2 summarizes the effect size of the DID estimate across all the quartiles (Online Table 9 provides the corresponding results for the rest of the quartiles). Whereas a collaboration gives an additional boost of +61.2% ($= e^{0.478} - 1$, p -value < 0.01) to the plays of artists in the first quartile (Model 2), the artists in the fourth quartile, that is, those who are considered the elite artists, experience a negative effect of -26.4% ($= e^{-0.306} - 1$, p -value < 0.01) from releasing a collaboration song (Model 3). This validates Hypothesis 4 and indicates an existence of the ceiling effect, whereby the payoff of a collaboration is much greater for the artists with lower economic capital. Note that this insight is also supported by the results of Section 5.3, in which we use the artist's economic capital as a moderator.

An additional exploration of considering the effect of multiple collaborations between the same pair of artists further reveals some interesting insights. We let $\text{Number_Collab}_{ik}$ be the number of times artist i and the pair in k have come together prior to the focal collaboration (zero for the first time they work together and positive otherwise). We find that an increase in the number of collaborations between the same pair has a negative impact on the total number of plays of the focal artist (Model 4). This may be attributable to the fact that the pairing loses its novelty after the first time and the audience perceives the subsequent collaborations between the same pair to be less exciting.

5.2. Focal Song and the Spillover Effects

The estimates for the effect on focal song plays are provided in Table 2 and Online Table 10. Model 5 shows a positive effect of +9.6% ($= e^{0.092} - 1$, p -value < 0.01) of the focal song being a collaboration effort on the number of plays it garners. This effect is especially noticeable for the elite artists at +41.6% ($= e^{0.348} - 1$, p -value < 0.01 , Figure 2). Hence the collaboration, on average, garners a larger number of plays than an individual song, and the lift is the highest for elite artists.

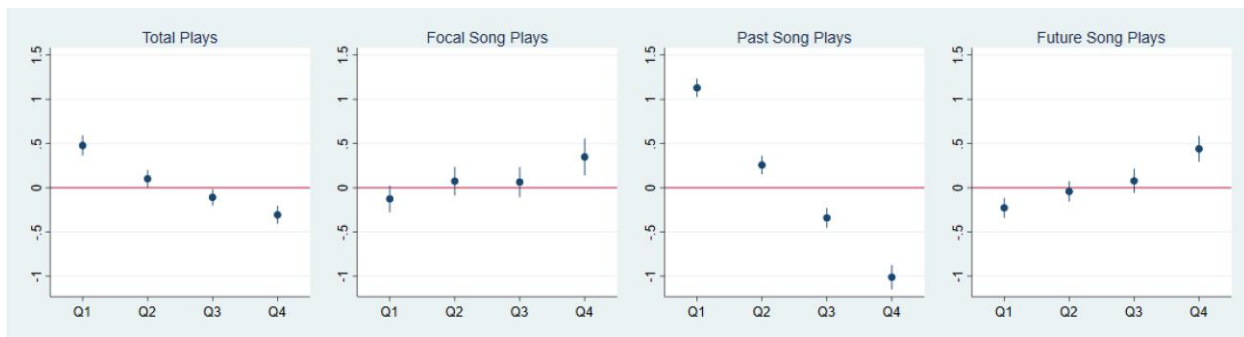
We now turn our attention to the spillover effects of this collaboration project. The effect on past songs in an artist's portfolio is insignificant across the full sample.

Table 2. Impact of a Collaboration in the Base Model (1)

	Total plays				Focal song plays				Past song plays				Future song plays			
	Full sample (1)	Q1 (2)	Q4 (3)	Full sample (4)	Full sample (5)	Q1 (6)	Q4 (7)	Full sample (8)	Full sample (9)	Q1 (10)	Q4 (11)	Full sample (12)	Full sample (13)	Q1 (14)	Q4 (15)	Full sample (16)
(Intercept)	−0.342*** (0.0939)	0.567 (0.363)	0.0955 (0.144)	−1.324*** (0.487)	−0.788 (0.684)	−2.074*** (0.365)	−0.442 (0.636)	2.426 (2.848)	−0.124*** (0.0458)	0.360 (0.288)	0.0929 (0.202)	−0.211 (0.275)	−3.612*** (0.138)	−1.893*** (0.264)	−4.367*** (0.284)	−2.692*** (0.225)
$AFTER_{\tau}$	0.0609*** (0.0187)	0.118*** (0.0409)	0.0358 (0.0384)	0.0611*** (0.0186)	1.094*** (0.0316)	1.019*** (0.0599)	1.289*** (0.0715)	1.094*** (0.0316)	−1.712*** (0.0244)	−1.579*** (0.0492)	−1.942*** (0.0506)	−1.712*** (0.0243)				
C_{ik}	−0.0205 (0.0349)	−1.586*** (0.0600)	1.583*** (0.0632)	0.0449 (0.0466)	−0.000502 (0.0285)	0.000404 (0.0479)	−0.00223 (0.0684)	0.0611 (0.0378)	−0.0230 (0.0383)	−1.582*** (0.0566)	1.570*** (0.0710)	0.0286 (0.0465)				
$AFTER_{\tau} \times C_{ik}$	0.0448* (0.0271)	0.478*** (0.0585)	−0.306*** (0.0523)	0.0947*** (0.0326)	0.0915** (0.0448)	−0.126 (0.0769)	0.348*** (0.106)	0.125** (0.0510)	0.0119 (0.0356)	1.129*** (0.0532)	−1.012*** (0.0699)	−0.0275 (0.0414)	0.0747*** (0.0232)	−0.227*** (0.0410)	0.440*** (0.0514)	0.101*** (0.0301)
$NumberCollab_{ik}$				−0.0674** (0.0319)				−0.0635*** (0.0243)				−0.0532* (0.0273)				−0.0272 (0.0175)
$AFTER_{\tau} \times C_{ik} \times NumberCollab_{ik}$				−0.0515*** (0.0179)				−0.0345 (0.0224)				0.0405* (0.0226)				
Artist controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pair-level fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of artist-songs	4,232	1,058	1,058	4,232	4,232	1,058	1,058	4,232	4,232	1,058	1,058	4,232	4,232	1,058	1,058	4,232
Number of observations	253,920	63,480	63,480	253,920	253,920	63,480	63,480	253,920	253,920	63,480	63,480	253,920	220,064	55,016	55,016	220,064
Adjusted R^2	0.453	0.470	0.542	0.455	0.370	0.334	0.412	0.371	0.602	0.589	0.731	0.602	0.353	0.302	0.424	0.353

Note. Standard errors in parentheses.
* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Figure 2. (Color online) Effect Size of the DID Estimate at 95% Confidence Intervals for Corresponding Plays Across Quartiles



A more nuanced look through Table 2 and Figure 2 (see Online Table 11 for full details) shows that the spillover effect is positive only for the artists in the first and second quartiles, those who were relatively less known and for whom the collaboration can bring the old catalogue to the attention of new listeners, supporting Hypothesis 2; in contrast, artists in the fourth quartile experience a drop of 63.6% ($= e^{-1.012} - 1$, p -value < 0.01), suggesting that their previously successful songs are more cannibalized by the collaboration project compared with a solo release. Collaboration, however, has a positive impact on the songs released in the future ($+7.7\% = e^{0.075} - 1$, p -value < 0.05), most of which is driven by the better performance of the songs released in the future by the collaborating artists in the fourth quartile in comparison with their matched controls ($+55.3\% = e^{0.440} - 1$, see Table 2, Online Table 12, and Figure 2). This validates Hypothesis 3 and suggests that a collaboration can be a mechanism to rejuvenate the career of an elite artist. Note that, because there are no observations for future song plays in the pretreatment period, $AFTER_{it}$ and C_{it} are dropped.

To summarize, elite artists benefit from the collaboration by increasing future performance at the expense of cannibalization of past songs, whereas nonelite artists experience increased attention for their previous works. Because the catalogue of past songs of any artist is much larger than the focal and future songs released in one year's time from the treatment period, the overall effect on the career of the artists measured in total plays is more in line with the effect observed for plays of the songs released in the past. See Online Appendix C.2, Table 18, for a summary of these results. Finally, this arrangement can be interpreted to be a direct result of the transfer of capital between the artists. In particular, the economic, social, and cultural distances between the artists are important drivers of the success of a collaboration. We isolate the effect of these moderators in the next section.

5.3. The Role of the Collaborator

Table 3 shows the combined effect of the moderating factors on the number of plays garnered by the focal

artist (see Online Tables 13–16 for details including breakdown onto focal, past, and future song plays). Note that, in this analysis, we control for the focal artist's capital through $Quartile_{ikt}$. We find that this control is important, in line with the stratified analysis of Section 5.1.

We first look at the effect of economic capital difference, measured by $\Delta CumPlays_{ikt}$, that is, the proportion of the plays between the artists in a dyad. This moderator shows a negative effect for total plays and past song plays (Models 17 and 21), which is in line with Hypothesis 5. From this, we can conclude that the boost received from a collaboration is much higher when the focal artist is a nonelite and the collaborating partner is higher than the artist in the status order as opposed to the focal artist being elite and the partner nonelite. However, it also shows a positive effect on the plays of the songs released in the future (Model 23): if an artist collaborates with someone with higher economic capital, then future songs are played less at the rate of 0.10% as the counterpart's plays increase by 1%. This is in line with our results from Models 11 and 15 in the main analysis: when a nonelite artist collaborates with an elite, it provides a boost to the past profile of the artist's career given the exposure lent by the elite partner. Note that the quadratic check reveals that this effect is concave increasing as $\Delta CumPlays_{ikt}$ increases. Model 21 shows that this exposure is increasing in the partner's economic capital. In contrast, songs from the artist's past receive much more attention than any of the artist's subsequent releases, thereby bringing the number of plays for the songs released in the future down and even more so when the partner's economic capital is larger as shown in Model 23. To summarize, Hypothesis 5 is generally validated (Model 17) although the choice of a partner with higher or lower economic capital changes the nature of spillover effects and affects each part of the portfolio differently (Models 21 and 23).

We now examine the effect of social capital differences: the effect of log-degree centrality difference on the number of plays garnered by the focal artist is linearly decreasing for focal song plays, that is, collaborating

Table 3. Effect of Complementarities in a Collaboration in Model (2)

	Total plays		Focal song plays		Past song plays		Future song plays	
	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)
(Intercept)	−0.0976 (0.851)	−0.0551 (0.894)	−1.029 (1.275)	−0.998 (1.308)	0.202 (0.509)	0.247 (0.558)	−2.011*** (0.435)	−2.088*** (0.514)
$AFTER_{\tau}$	0.0657* (0.0372)	0.0648* (0.0373)	1.094*** (0.0619)	1.094*** (0.0619)	−1.712*** (0.0456)	−1.716*** (0.0456)		
C_{ik}		0.168*** (0.0628)	0.307*** (0.0598)	0.00874 (0.0876)	0.161* (0.0673)	0.162** (0.0920)	0.133	
$AFTER_{\tau} \times C_{ik}$	0.0695 (0.0520)	0.0335 (0.0725)	0.363*** (0.0943)	0.266** (0.124)	−0.0811 (0.0638)	−0.121 (0.0855)	0.238*** (0.0448)	0.426*** (0.0710)
$AFTER_{\tau} \times C_{ik} \times \Delta CumPlays_{ik\tau}$	−0.0642*** (0.0223)	−0.0658*** (0.0219)	0.0405 (0.0498)	0.0310 (0.0480)	−0.311*** (0.0275)	−0.306*** (0.0268)	0.102*** (0.0289)	0.104*** (0.0283)
$AFTER_{\tau} \times C_{ik} \times (\Delta CumPlays_{ik\tau})^2$		0.00319 (0.0109)		0.0139 (0.0205)		0.0000328 (0.0117)		−0.0480*** (0.0124)
$AFTER_{\tau} \times C_{ik} \times \Delta DegreeCentrality_{ik\tau}$	0.00266 (0.0244)	0.00726 (0.0243)	−0.110** (0.0531)	−0.0712 (0.0526)	0.0945*** (0.0274)	0.0783*** (0.0265)	0.0281 (0.0290)	0.0125 (0.0321)
$AFTER_{\tau} \times C_{ik} \times (\Delta DegreeCentrality_{ik\tau})^2$		0.0120 (0.0100)		0.0487** (0.0207)		−0.0103 (0.0102)		−0.0230* (0.0121)
$AFTER_{\tau} \times C_{ik} \times \Delta Art_{ik\tau}$	2.633 (20.96)	32.18 (62.90)	6.130 (51.56)	−3.015 (88.94)	49.62** (23.54)	46.22 (65.62)	−20.00 (18.18)	−55.15 (62.54)
$AFTER_{\tau} \times C_{ik} \times (\Delta Art_{ik\tau})^2$		−1953.3 (3231.5)		599.4 (4735.1)		451.1 (3484.2)		2139.4 (3147.8)
$AFTER_{\tau} \times C_{ik} \times \Delta Origin_{ik\tau}$	0.896 (0.679)	0.569 (1.432)	2.284* (1.203)	5.202* (2.894)	0.401 (0.796)	−3.688** (1.678)	−1.100 (0.750)	0.640 (1.840)
$AFTER_{\tau} \times C_{ik} \times (\Delta Origin_{ik\tau})^2$		1.875 (8.964)		−21.69 (15.17)		28.93*** (9.325)		−11.28 (9.716)
Artist controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pair-level fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of artist-songs	1,148	1,148	1,148	1,148	1,148	1,148	1,148	1,148
Number of observations	68,878	68,878	68,878	68,878	68,878	68,878	59,696	59,696
Adjusted R^2	0.509	0.513	0.394	0.398	0.664	0.667	0.395	0.399

Notes. Standard errors in parentheses. Two-way interactions are excluded for the sake of brevity.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

with someone with higher social capital increases plays. The variation remains significant with a 1% increase in the value of the moderator causing focal song plays to change by -0.10% (Model 19). Collaborating with an artist who is socially well connected, hence, grants the focal artist access to the partner's collaborators, thereby providing the artist with better exposure and broader distribution (Van den Bulte and Moenaert 1998, Ductor 2015) to monetize the social tie created by the collaboration. Moreover, the transfer of social capital is larger when the difference in social capital increases, validating Hypothesis 6 for social capital in the case of the focal song. On the other hand, collaborating with someone with a high social capital does not necessarily increase the plays of the past songs (Model 21). This may be because the focal song commands so much attention from audiences that the plays of the past songs get cannibalized. Overall, transfer of social capital is apparent when we look at the focal song, but there is no evidence of it in the rest of the artist's portfolio, implying limited spillovers in relation to social capital.

Finally, we explore the role of cultural distance. The results indicate the relationship between the distance in

artistic attributes and the plays garnered by the focal artist mostly remains insignificant. We only see a slightly positive moderating impact of the distance in artistic attributes on the plays of the past songs. Indeed, a combination of two extremely different styles of music is more likely to produce a focal song that is so unique compared with mainstream genres that, barring a certain fraction of the audience, is rejected by the majority (Mark 1998). This drives audiences to the past profile of the focal artist to seek solace in its familiar content. Hypothesis 7, with respect to artistic distance, hence, remains invalidated. On the other hand, the effect of the distance in origin-culture attributes linearly increases focal song plays (Model 19) implying that the impact of collaboration largely increases with the origin-culture distance between artists. This suggests that artists coming from extremely different cultures create a buzz of excitement among audiences, which makes the focal song perform very well in the market. The estimates for the moderation effect of origin-culture distance in the case of future song plays indicate an inverted U-shaped effect (Model 24). However, this effect is statistically insignificant. Nevertheless, this still has a negative impact

on the songs from the past, hence producing a U-shaped effect (Model 22). Hence, Hypothesis 7 remains invalidated in the case of cultural origin of the collaborating artists as well.

A summary of the complementarities identified in this analysis is given in Online Appendix C.2, Table 19, and the robustness checks carried out to test the corresponding curvilinear effects are given in Online Appendix D. Further research can be conducted to identify which metrics of diversity are more important than others and to isolate the factors that are more valued by the audiences for more informed collaboration-related decisions made by the artists.

6. Additional Robustness Checks

In this section, we outline the additional robustness checks that we carry out to address several conceptual and econometric concerns that arise in our effort to isolate the hypothesized relationships.

6.1. Potential Cannibalization Effects Across Treatment and Control

An important requirement for a valid application of the DID methodology is that the treatment or the lack thereof should not affect the other group in any way. In our context, it is essential to compare artists who release a collaboration or a solo song in the same week in order to make sure that they both face the same market dynamics. Hence, it is possible for the collaboration and solo songs that are identified to be a match through our matching procedure to have a cannibalization effect on each other because they are available for airing on all the radio stations in the data set.

To check for this, we first go back to our matching procedure in Section 4.2.1 to get an additional match of an artist who has a similar career trajectory as that of the artist who releases a collaboration song in terms of career age ($CareerAge_{it}$), new songs released in the recent past ($IndivSongs_{it}$ and $CollabSongs_{it}$), the record label the artist was working with for the artist's last release ($PrevBigLabel_{it}$), and the frequency with which the artist's songs are played on the radio ($CumPlays_{it}$). By using a caliper of 0.5, we are able to get 1,688 groups of matched artists—each group k containing three artists: one who releases a collaboration song, one who is with a solo song, and one without a release—all of which are comparable to each other along the matching dimensions utilized.

We then carry out two checks as explained in Online Appendix E.1, in which the effect of a release of song in the posttreatment period is examined through a within model and a stricter approach of triple differences (difference-in-difference-in-differences) used in, for example, Fisher et al. (2018) to isolate the effect of releasing a collaboration song in comparison with a solo song and

no release. The first check is instrumental in showing that the release of any kind of song does not cannibalize the career trajectory of artists without a new release. The second check further validates this with the resultant effect of collaboration being significant, positive, and almost identical to our results in the main analysis, accounting for both the decisions of releasing a song and releasing a collaboration versus a solo song. This provides evidence of the fact that our matching procedure is instrumental in accounting for not only the non-random decision of releasing a collaboration song, but it also corrects for the self-selection bias arising because of the decision of introducing a song at a certain period in time.

6.2. Effect of Treatment Across Time

We adopt the estimation procedure followed by Autor (2003) in the context of a DID approach. Whereas the matching statistics from Online Table 8 provide evidence for an efficient balance between the treatment and control groups, this allows us to formally test whether there is a significant difference in the values of the dependent variable between the two groups prior to the treatment through this method. Additionally, it also helps us to flesh out the weekly effect of a collaboration on the dependent variables (Calvo et al. 2020).

The details of this check are given in Online Appendix E.2. Online Figure 6 illustrates the week-level effects described by Online Equation (3) and corroborates that the treatment effect is significant only after the treatment time (Online Appendix E.2, Table 23).

6.3. Alternate Variable Specifications and Matching Parameters

Throughout our analysis, our dependent variables of plays recorded on the radio are normalized to the number of songs an artist has in the artist's catalogue. We extend this analysis by using simple sums instead, which provides effect sizes of the hypothesized relationships to be qualitatively similar to our main analysis (Online Appendix E.3). We also extend our sample from the top 90th percentile artists (who garner 91% of plays recorded on radio stations) down to a sample of artists in the 40th percentile. We see that the positive effect of releasing a collaboration versus a solo song is not observable in the 70th to 90th percentile analysis but significantly affects the profiles of artists in the 40th to 70th percentiles (Online Appendix E.4). Moreover, the impact on focal, past, and future songs remains qualitatively similar in sign. This indicates that our hypothesized relationships differ in salience, that is, may become weaker or stronger based on the varied positions artists occupy in the music industry. We also find that releasing a song with a big record label does not give a disproportionate advantage to the artists

who release a collaboration song (Online Appendix E.5).

In our main analysis, we use a caliper of 0.5 to get 2,116 matches. We alter this parameter to several different levels to get results similar to our main specification; two of them, with caliper of 0.15 and 0.75, are included in Online Appendix E.6.1. An important exclusion in our matching strategy is that of not considering the artistic and cultural attributes of the artists because of the lack of availability of the corresponding data (only available for artists in the higher percentiles). Nevertheless, using the much-reduced subsample of artists for whom these attributes are available, we augment our matching procedure with these factors but remove the caliper to get only 1,111 matches. As shown in Online Appendix E.6.2, this analysis with additional matching parameters but a slacker control on the quality of matches yields results similar to our main findings. Finally, we use coarsened exact matching and Mahalanobis distance-based matching as robustness checks for the propensity score matching. The former provides weighted matches (2,521 pairs), whereas the latter provides the nearest matches based on individual calipers (3,746 pairs). We see in Online Appendices E.6.3 and E.6.4 that these alternate matching methodologies also show that collaboration provides a positive boost to an artist's career in comparison with a solo release.

7. Conclusion

Collaborative projects in music are an effective tool for artists to boost their careers. Indeed, we find in our study that the effect of a collaboration is not just limited to the focal project. Spillover effects are observed, affecting the plays of past and future songs. Specifically, we find that the total profile of an artist who collaborates receives a significantly larger boost compared with those who do not collaborate. We find evidence of a ceiling effect in which elite artists have much less to gain from a collaboration project compared with nonelites. Moreover, we find that complementarities are important. High economic and social capital artists are indeed the ideal partners when seeking collaborators, transferring their capital to the nonelite focal artist. On the other hand, we find little evidence that artistic and cultural distances have an impact on the collaboration.

7.1. Implications for the Music Industry

Drawing on our findings, we illustrate how a data-driven model rooted in fundamental theoretical constructs can be used to provide decision support for artists in their search for potential collaborators. The counterfactual analysis detailed in Online Appendix F provides a prediction of how Shakira's career trajectory would evolve under different scenarios of releasing a solo song in comparison with collaborating with any of

her select few peers. In this specific example, our framework helps in identifying Eros Ramazzotti to be the best choice for her to collaborate with to ensure the maximum payoff. However, the overall exercise is instrumental in underscoring the fact that artists can be more strategic when choosing their potential partners by taking into account the relative positioning of the options available to them with respect to their own spot in the artistic and cultural sphere of the industry. In an age that is increasingly becoming Asimovian, in which artificial intelligence is acclaimed for being able to autonomously create high-quality cultural content (Moura 2007) and a society's need for human cultural creatives may be threatened, our work can be interpreted as a data-driven empowering tool for artists, who can receive quantitative decision support to choose the right collaborators, with the right complementarities, at the right time.

7.2. Implications for the Cultural Industries

Whereas this paper focuses on music, the fundamental elements of our theoretical framework can be adopted to study other forms of collaborative NPD efforts within the domain of cultural production. For example, when superstar actors are cast for a part in a particular movie, along with increasing the probability of the movie becoming a success (Elberse 2007), their involvement can also have positive spillover effects on the careers of their costars. Whereas it is not common for artists to collaborate in the production of paintings or sculptures, they may certainly increase the visibility of their artworks by participating in multiple-artist exhibitions, which are bound to draw in more crowds. The artist-level findings at the microlevel may be extended to the production of cultural products that involve larger team efforts at the meso- and macro-levels as well (Roels and Staats 2021). For example, *Rolling Stone* magazine primarily focuses on meticulous reporting on music, entertainment, and politics, whereas *Sports Illustrated* publishes articles that are exclusively sports-related. These two extremely popular magazine outlets may collaborate on a special edition and leverage their respective skills and resources to produce a volume that not only provides comprehensive coverage of pop culture, but also extends their base market by enticing the usual readers of their partner into subscribing to their regular publications. Hence, our theoretical framework can prove to be a starting point for carrying out more comprehensive exploration of such alliances.

7.3. Limitations and Future Work

Of course, this paper has some limitations that also pose as excellent ventures for future research. First, by aggregating our results to a pan-European level, we miss the opportunity to explore the variance present at the country level. Consequently, we do not identify the dynamics of the effect of the collaboration in the countries of origin

of the two collaborating artists, which may provide interesting insights. Second, we have limited our attention to top artists (90th percentile in the main model, 40th percentile as robustness check) with an aim to focus on the commercial music industry. With the increased influence of streaming services such as Spotify and YouTube on today's media consumption, the lines between the commercial and noncommercial music are becoming increasingly blurred (Wlömert and Papies 2016). It would be interesting to measure the cross-media impact of collaborations, for example, what was the impact of Shakira's appearance in cover artist Daniela Andrade's video of "Hips Don't Lie" on the number of subscribers to the latter artist's YouTube channel. Third, we currently do not measure the sustained learning effect of the collaboration (Singh 2005, Liu et al. 2018) or quantify the increase or decrease in the innovation potential of the artists (Mauskapf et al. 2017), which is a natural extension of our results. Fourth, we assume symmetric investments in terms of costs and efforts toward the NPD project, keeping the revenue sharing out of scope in this work (unobservable in our data). It would be interesting to see how these resource-sharing and allocation mechanisms, as discussed in collaborative and hierarchical contexts by Bhaskaran and Krishnan (2009), Mihm et al. (2010), and Hutchison-Krupat and Kavadias (2015), fare in the music industry. Finally, prescriptive models that particularly focus on identifying optimal complementarities can be developed based on our theoretical framework. To conclude, we believe that the management of cultural operations, including design, production, and distribution, can generally benefit from quantitative, data-driven approaches grounded in classic theories, such as the one developed in this paper.

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